where we view $\{x_k \mid k = 0, \ldots, N-1\}$ and $\{u_k \mid k = 0, \ldots, N-1\}$ as state and control trajectories, respectively, which approximate the corresponding continuous-time trajectories:

$$x_k \approx x(k\delta), \qquad u_k \approx u(k\delta).$$

We also replace the cost function (2.56) by

$$g_N(x_N) + \sum_{k=0}^{N-1} \delta \cdot g_k(x_k, u_k),$$

where

$$g_N(x_N) = G\big(x(N\delta)\big), \qquad g_k(x_k, u_k) = g\big(x(k\delta), u(k\delta), k\delta\big).$$

Thus the approximation in value space scheme with time discretization takes the form

$$\tilde{\mu}_k(x_k) \in \arg\min_{u_k \in U} \Big[\delta \cdot g_k(x_k, u_k) + \tilde{J}_{k+1}\big(x_k + \delta \cdot h_k(x_k, u_k)\big)\Big]; \qquad (2.57)$$

where $\tilde{J}_{k+1}$ is the function that approximates the cost-to-go starting from a state at time $k+1$. We note here that the ratio of the terms $\delta \cdot g_k(x_k, u_k)$ and $\tilde{J}_{k+1}\big(x_k + \delta \cdot h_k(x_k, u_k)\big)$ is likely to tend to 0 as $\delta \to 0$, since $\tilde{J}_{k+1}\big(x_k + \delta \cdot h_k(x_k, u_k)\big)$ ordinarily stays roughly constant at a nonzero level as $\delta \to 0$. This suggests that the one-step lookahead minimization may be degraded substantially by discretization, and other errors, including rollout truncation and terminal cost approximation. Note that a similar sensitivity to errors may occur in other discrete-time models that involve frequent selection of decisions, with cost per stage that is very small relative to the cumulative cost over many stages and/or the terminal cost.

To deal with this difficulty, we subtract the constant $\tilde{J}_k(x_k)$ in the one-step-lookahead minimization (2.57), and write

$$\tilde{\mu}_k(x_k) \in \arg\min_{u_k \in U} \Big[\delta \cdot g_k(x_k, u_k) + \Big(\tilde{J}_{k+1}\big(x_k + \delta \cdot h_k(x_k, u_k)\big) - \tilde{J}_k(x_k)\Big)\Big]; \qquad (2.58)$$

since $\tilde{J}_k(x_k)$ does not depend on u_k, the results of the minimization are not affected. Assuming $\tilde{J}_k$ is differentiable with respect to its argument, we can write

$$\tilde{J}_{k+1}\big(x_k + \delta \cdot h_k(x_k, u_k)\big) - \tilde{J}_k(x_k) \approx \delta \cdot \nabla_x \tilde{J}_k(x_k)' h_k(x_k, u_k),$$

where $\nabla_x \tilde{J}_k$ denotes the gradient of J_k (a column vector), and prime denotes transposition. By dividing with δ, and taking informally the limit as $\delta \to 0$, we can write the one-step lookahead minimization (2.58) as

$$\tilde{\mu}(t) \in \arg\min_{u(t) \in U} \Big[g\big(x(t), u(t), t\big) + \nabla_x \tilde{J}_t\big(x(t)\big)' h\big(x(t), u(t), t\big)\Big], \qquad (2.59)$$

where $\tilde{J}_t(x)$ is the continuous-time cost function approximation and $\nabla_x \tilde{J}_t(x)$ is its gradient with respect to x. This is the correct analog of the approximation in value space scheme (2.54) for continuous-time problems.

Rollout for Continuous-Time Optimal Control

In view of the value approximation scheme of Eq. (2.59), it is natural to speculate that the continuous-time analog of rollout with a base policy of the form

$$\pi = \Big\{\mu_t\big(x(t)\big) \mid 0 \le t \le T\Big\}, \tag{2.60}$$

where $\mu_t\big(x(t)\big) \in U$ for all $x(t)$ and t, has the form

$$\tilde{\mu}_t\big(x(t)\big) \in \arg\min_{u(t)\in U} \Big[g\big(x(t), u(t), t\big) + \nabla_x J_{\pi,t}\big(x(t)\big)' h\big(x(t), u(t), t\big)\Big]. \tag{2.61}$$

Here $J_{\pi,t}\big(x(t)\big)$ is the cost of the base policy π starting from state $x(t)$ at time t, and satisfies the terminal condition

$$J_{\pi,T}\big(x(T)\big) = G\big(x(T)\big).$$

Computationally, the inner product in the right-hand side of the above minimization can be approximated using the finite difference formula

$$\nabla_x J_{\pi,t}\big(x(t)\big)' h\big(x(t), u(t), t\big) \approx \frac{J_{\pi,t}\Big(x(t) + \delta \cdot h\big(x(t), u(t), t\big)\Big) - J_{\pi,t}\big(x(t)\big)}{\delta},$$

which can be calculated by running the base policy π starting from $x(t)$ and from $x(t) + \delta \cdot h\big(x(t), u(t), t\big)$. (This finite differencing operation may involve tricky computational issues, but we will not get into this.)

An important question is how to select the base policy π. A choice that is often sensible and convenient is to choose π to be a "short-sighted" policy, which takes into account the "short term" cost from the current state (say for a very small horizon starting from the current time t), but ignores the remaining cost. An extreme case is the *myopic* policy, given by

$$\mu_t\big(x(t)\big) \in \arg\min_{u\in U} g\big(x(t), u(t), t\big).$$

This policy is the continuous-time analog of the greedy policy that we discussed in the context of discrete-time problems, and the traveling salesman Example 1.2.3 in particular.

The following example illustrates the rollout algorithm (2.61) with a problem that has a special property: the base policy cost $J_{\pi,t}\big(x(t)\big)$ is independent of $x(t)$ (it depends only on t), so that $\nabla_x J_{\pi,t}\big(x(t)\big) \equiv 0$. In this case, in view of Eq. (2.59), the rollout policy is myopic. It turns out that the optimal policy in this example is also myopic, so that the rollout policy is optimal, even though the base policy is very poor.

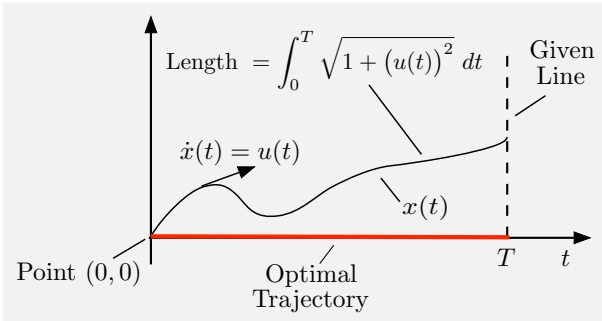

Figure 2.6.1 Problem of finding a curve of minimum length from a given point to a given line, and its formulation as a calculus of variations problem.

Example 2.6.1 (A Calculus of Variations Problem)

This is a simple example from the classical context of calculus of variations (see [Ber17a], Example 7.1.3). The problem is to find a minimum length curve that starts at a given point and ends at a given line. Without loss of generality, let $(0, 0)$ be the given point, and let the given line be the vertical line that passes through $(T, 0)$, as shown in Fig. 2.6.1.

Let $\big(t, x(t)\big)$ be the points of the curve, where $0 \le t \le T$. The portion of the curve joining the points $\big(t, x(t)\big)$ and $\big(t + dt, x(t + dt)\big)$ can be approximated, for small dt, by the hypotenuse of a right triangle with sides dt and $\dot{x}(t)dt$. Thus the length of this portion is

$$\sqrt{(dt)^2 + \big(\dot{x}(t)\big)^2 (dt)^2},$$

which is equal to

$$\sqrt{1 + \big(\dot{x}(t)\big)^2}\, dt.$$

The length of the entire curve is the integral over $[0, T]$ of this expression, so the problem is to

$$\begin{aligned} &\text{minimize} \quad \int_0^T \sqrt{1 + \big(\dot{x}(t)\big)^2}\, dt \\ &\text{subject to} \quad x(0) = 0. \end{aligned}$$

To reformulate the problem as a continuous-time optimal control problem, we introduce a control u and the system equation

$$\dot{x}(t) = u(t), \qquad x(0) = 0.$$

Our problem then takes the form

$$\text{minimize} \quad \int_0^T \sqrt{1 + \big(u(t)\big)^2}\, dt.$$

This is a problem that fits our continuous-time optimal control framework, with

$$h\big(x(t),u(t),t\big) = u(t), \quad g\big(x(t),u(t),t\big) = \sqrt{1+\big(u(t)\big)^2}, \quad G\big(x(T)\big) = 0.$$

Consider now a base policy π whereby the control depends only on t and not on x. Such a policy has the form

$$\mu_t\big(x(t)\big) = \beta(t), \qquad \text{for all } x(t),$$

where $\beta(t)$ is some scalar function. For example, $\beta(t)$ may be constant, $\beta(t) \equiv \bar{\beta}$ for some scalar $\bar{\beta}$, which yields a straight line trajectory that starts at $(0,0)$ and makes an angle ϕ with the horizontal with $\tan(\phi) = \bar{\beta}$. The cost function of the base policy is

$$J_{\pi,t}\big(x(t)\big) = \int_t^T \sqrt{1+\beta(\tau)^2}\, d\tau,$$

which is independent of $x(t)$, so that $\nabla_x J_{\pi,t}\big(x(t)\big) \equiv 0$. Thus, from the minimization of Eq. (2.61), we have

$$\tilde{\mu}_t\big(x(t)\big) \in \arg\min_{u(t)\in\Re} \sqrt{1+\big(u(t)\big)^2},$$

and the rollout policy is

$$\tilde{\mu}_t\big(x(t)\big) \equiv 0.$$

This is the optimal policy: it corresponds to the horizontal straight line that starts at $(0,0)$ and ends at $(T,0)$.

Rollout with General Base Heuristics - Sequential Improvement

An extension of the rollout algorithm (2.61) is to use a more general base heuristic whose cost function $H_t\big(x(t)\big)$ can be evaluated by simulation. This rollout algorithm has the form

$$\tilde{\mu}(t) \in \arg\min_{u(t)\in U} \Big[g\big(x(t),u(t),t\big) + \nabla_x H_t\big(x(t)\big)' h\big(x(t),u(t),t\big)\Big].$$

Here the policy cost function $J_{\pi,t}$ is replaced by a more general differentiable function H_t, obtainable through a base heuristic, which may lack the sequential consistency property that is inherent in policies.

We will now show a cost improvement property of the rollout algorithm based on the natural condition

$$H_T\big(\tilde{x}(T)\big) = G\big(\tilde{x}(T)\big), \tag{2.62}$$

and the assumption

$$\min_{u(t)\in U}\Big[g\big(x(t),u(t),t\big)+\nabla_t H_t\big(x(t)\big)+\nabla_x H_t\big(x(t)\big)'h\big(x(t),u(t),t\big)\Big]\le 0, \tag{2.63}$$

for all $\big(x(t),t\big)$, where $\nabla_x H_t$ denotes gradient with respect to x, and $\nabla_t H_t$ denotes gradient with respect to t. This assumption is the continuous-time analog of the sequential improvement condition of Definition 2.3.2 [cf. Eq. (2.13)]. Under this assumption, we will show that

$$J_{\tilde\pi,0}\big(x(0)\big)\le H_0\big(x(0)\big), \tag{2.64}$$

i.e., the cost of the rollout policy starting from the initial state $x(0)$ is no worse than the base heuristic cost starting from the same initial state.

Indeed, let $\big\{\tilde x(t)\mid t\in[0,T]\big\}$ and $\big\{\tilde u(t)\mid t\in[0,T]\big\}$ be the state and control trajectories generated by the rollout policy starting from $x(0)$. Then the sequential improvement condition (2.63) yields

$$g\big(\tilde x(t),\tilde u(t),t\big)+\nabla_t H_t\big(\tilde x(t)\big)+\nabla_x H_t\big(\tilde x(t)\big)'h\big(\tilde x(t),\tilde u(t),t\big)\le 0$$

for all t, and by integration over $[0,T]$, we obtain

$$\int_0^T g\big(\tilde x(t),\tilde u(t),t\big)\,dt+\int_0^T\Big(\nabla_t H_t\big(\tilde x(t)\big)+\nabla_x H_t\big(\tilde x(t)\big)'h\big(\tilde x(t),\tilde u(t),t\big)\Big)\,dt\le 0. \tag{2.65}$$

The second integral above can be written as

$$\begin{aligned}\int_0^T&\Big(\nabla_t H_t\big(\tilde x(t)\big)+\nabla_x H_t\big(\tilde x(t)\big)'h\big(\tilde x(t),\tilde u(t),t\big)\Big)\,dt\\&=\int_0^T\Big(\nabla_t H_t\big(\tilde x(t)\big)+\nabla_x H_t\big(\tilde x(t)\big)'\frac{d\tilde x(t)}{dt}\Big)\,dt,\end{aligned}$$

and its integrand is the total differential with respect to time: $\frac{d}{dt}\Big(H_t\big(\tilde x(t)\big)\Big)$. Thus we obtain from Eq. (2.65)

$$\begin{aligned}\int_0^T g\big(\tilde x(t),\tilde u(t),t\big)\,dt&+\int_0^T\frac{d}{dt}\Big(H_t\big(\tilde x(t)\big)\Big)\,dt\\&=\int_0^T g\big(\tilde x(t),\tilde u(t),t\big)\,dt+H_T\big(\tilde x(T)\big)-H_0\big(\tilde x(0)\big)\le 0.\end{aligned} \tag{2.66}$$

Since

$$H_T\big(\tilde x(T)\big)=G\big(\tilde x(T)\big),$$

[cf. Eq. (2.62)] and $\tilde x(0)=x(0)$, from Eq. (2.66) [which is a direct consequence of the sequential improvement condition (2.63)], it follows that

$$J_{\tilde\pi,0}\big(x(0)\big)=\int_0^T g\big(\tilde x(t),\tilde u(t),t\big)\,dt+G\big(\tilde x(T)\big)\le H_0\big(x(0)\big),$$

thus proving the cost improvement property (2.64).

Note that the sequential improvement condition (2.63) is satisfied if H_t is the cost function $J_{\pi,t}$ corresponding to a base policy π. The reason is that for any policy $\pi = \big\{\mu_t(x(t)) \mid 0 \le t \le T\big\}$ [cf. Eq. (2.60)], the analog of the DP algorithm (under the requisite mathematical conditions) is

$$0 = g\big(x(t), \mu_t(x(t)), t\big) + \nabla_t J_{\pi,t}\big(x(t)\big) + \nabla_x J_{\pi,t}\big(x(t)\big)' h\big(x(t), \mu_t(x(t)), t\big). \tag{2.67}$$

In continuous-time optimal control theory, this is known as the *Hamilton-Jacobi-Bellman equation*. It is a partial differential equation, which may be viewed as the continuous-time analog of the DP algorithm for a single policy; there is also a Hamilton-Jacobi-Bellman equation for the optimal cost function $J_t^*\big(x(t)\big)$ (see optimal control textbook accounts, such as [Ber17a], Section 7.2, and the references cited there). As illustration, the reader may verify that the cost function of the base policy used in the calculus of variations problem of Example 2.6.1 satisfies this equation. It can be seen from the Hamilton-Jacobi-Bellman Eq. (2.67) that when $H_t = J_{\pi,t}$, the sequential improvement condition (2.63) and the cost improvement property (2.64) hold.

Approximating Cost Function Differences

The preceding analysis suggests that when dealing with a discrete-time problem with a long horizon N, a system equation $x_{k+1} = f_k(x_k, u_k)$, and a small cost per stage $g_k(x_k, u_k)$ relative to the optimal cost-to-go function $J_{k+1}^*\big(f_k(x_k, u_k)\big)$, it is worth considering an alternative implementation of the approximation in value space scheme. In particular, we should consider approximating the cost differences

$$D_k^*(x_k, u_k) = J_{k+1}^*\big(f_k(x_k, u_k)\big) - J_k^*(x_k)$$

instead of approximating the optimal cost-to-go functions $J_{k+1}^*\big(f_k(x_k, u_k)\big)$. The one-step-lookahead minimization (2.54) should then be replaced by

$$\tilde{\mu}_k(x_k) \in \arg \min_{u_k \in U_k(x_k)} \big[g_k(x_k, u_k) + \tilde{D}_k(x_k, u_k)\big],$$

where $\tilde{D}_k$ is the approximation to D_k^*.

Note also that while for continuous-time problems, the idea of approximating the gradient of the optimal cost function is essential and comes out naturally from the analysis, for discrete-time problems, approximating cost-to-go differences rather than cost functions is optional and should be considered in the context of a given problem, possibly in conjunction with increased lookahead. Methods along this line include advantage updating, cost shaping, biased aggregation, and the use of baselines, for which we refer to the books [BeT96], [Ber19a], and [Ber20a]. A special method to explicitly approximate cost function differences is *differential training*, which was proposed in the author's paper [Ber97b], and was also discussed in Section 4.3.4 of the book [Ber20a].

The Case of Zero Cost per Stage

The most extreme and challenging case of small stage costs arises when the cost per stage is zero for all states, while a nonzero cost may be incurred only at termination. This type of cost structure occurs, among others, in games such as chess and backgammon (another interesting context is solving the Rubik's cube [AMS19], [MAS19]). It also occurs in several other contexts, including constraint programming problems (Section 2.1), where there is not even a terminal cost, just constraints to be satisfied.

Under these circumstances, the idea of approximating cost-to-go differences that we have just discussed may not be effective, and applying approximation in value space may involve serious challenges. An advisable remedy is to resort to long lookahead, either through multistep lookahead minimization (possibly augmented by pruning or incremental rollout ideas; cf. Section 2.4), or through some form of truncated rollout.

It may also be important to introduce a terminal cost function approximation by using problem simplification (solving a simpler problem, in place of the original), or neural network training. Chess, Go, backgammon, and other games are prime examples of zero cost per stage problems, where the use of a terminal cost function approximation, obtained through a neural network training, has been a critical part of the solution methodology.

2.7 APPROXIMATION IN VALUE SPACE - STOCHASTIC PROBLEMS

We will now discuss approximation in value space for stochastic problems. We first focus on rollout algorithms for finite state, control, and disturbance spaces. We will restrict ourselves to the case where the base heuristic is a policy $\pi = \{\mu_0, \ldots, \mu_{N-1}\}$. The rollout policy applies at state x_k the control $\tilde{\mu}_k(x_k)$ given by the minimization

$$\tilde{\mu}_k(x_k) \in \arg \min_{u_k \in U_k(x_k)} E\Big\{g_k(x_k, u_k, w_k) + J_{k+1,\pi}\big(f_k(x_k, u_k, w_k)\big)\Big\}.$$

Equivalently, the rollout policy $\tilde{\pi} = \{\tilde{\mu}_0, \ldots, \tilde{\mu}_{N-1}\}$ is obtained by minimization over the Q-factors $Q_{k,\pi}(x_k, u_k)$ of the base policy:

$$\tilde{\mu}_k(x_k) \in \arg \min_{u_k \in U_k(x_k)} Q_{k,\pi}(x_k, u_k),$$

where

$$Q_{k,\pi}(x_k, u_k) = E\Big\{g_k(x_k, u_k, w_k) + J_{k+1,\pi}\big(f_k(x_k, u_k, w_k)\big)\Big\}.$$

We first establish that the cost improvement property that we showed for deterministic problems under the sequential consistency condition carries through for stochastic problems. In particular, let us denote by $J_{k,\pi}(x_k)$

the cost corresponding to starting the base policy at state x_k, and by $J_{k,\tilde{\pi}}(x_k)$ the cost corresponding to starting the rollout algorithm at state x_k. We claim that

$$J_{k,\tilde{\pi}}(x_k) \leq J_{k,\pi}(x_k), \qquad \text{for all } x_k \text{ and } k. \tag{2.68}$$

We prove this inequality by induction similar to the deterministic case [cf. Eq. (2.12)]. Clearly it holds for $k = N$, since

$$J_{N,\tilde{\pi}} = J_{N,\pi} = g_N.$$

Assuming that it holds for index $k+1$, we have for all x_k,

$$\begin{aligned} J_{k,\tilde{\pi}}(x_k) &= E\Big\{g_k\big(x_k, \tilde{\mu}_k(x_k), w_k\big) + J_{k+1,\tilde{\pi}}\Big(f_k\big(x_k, \tilde{\mu}_k(x_k), w_k\big)\Big)\Big\} \\ &\leq E\Big\{g_k\big(x_k, \tilde{\mu}_k(x_k), w_k\big) + J_{k+1,\pi}\Big(f_k\big(x_k, \tilde{\mu}_k(x_k), w_k\big)\Big)\Big\} \\ &= \min_{u_k \in U_k(x_k)} E\Big\{g_k(x_k, u_k, w_k) + J_{k+1,\pi}\big(f_k(x_k, u_k, w_k)\big)\Big\} \\ &\leq E\Big\{g_k\big(x_k, \mu_k(x_k), w_k\big) + J_{k+1,\pi}\Big(f_k\big(x_k, \mu_k(x_k), w_k\big)\Big)\Big\} \\ &= J_{k,\pi}(x_k), \end{aligned} \tag{2.69}$$

where:

(a) The first equality is the DP equation for the rollout policy $\tilde{\pi}$.

(b) The first inequality holds by the induction hypothesis.

(c) The second equality holds by the definition of the rollout algorithm.

(d) The final equality is the DP equation for the base policy π.

The induction proof of the cost improvement property is thus complete.

The preceding cost improvement argument assumes that the cost functions $J_{k+1,\pi}$ of the base policy are calculated exactly. In practice, truncated rollout with terminal cost function approximation and limited simulation may be used to approximate $J_{k+1,\pi}$. In this case the cost function of the rollout policy can still be viewed as the result of a Newton step in the context of an approximation in value space scheme. Moreover, the cost improvement property can still be proved under some conditions that we will not discuss in this book; see the books [Ber12], [Ber19a], and [Ber20a].

Some Rollout Examples

Similar to deterministic problems, it has been observed empirically that for stochastic problems the rollout policy not only does not deteriorate

the performance of the base policy, but also typically produces substantial cost improvement, thanks to its underlying Newton step; see also the case studies referenced at the end of the chapter. To emphasize this point, we provide here an example of a nontrivial optimal stopping problem where the rollout policy is actually optimal, despite the fact that the base policy is rather naive. Such behavior is of course special and nontypical, but highlights the nature of the cost improvement property of rollout.

Example 2.7.1 (Optimal Stopping and Rollout Optimality)

Optimal stopping problems are characterized by the availability, at each state, of a control that stops the evolution of the system. We will consider a problem with two control choices: at each stage we observe the current state of the system and decide whether to continue or to stop the process. We formulate this as an N-stage problem where stopping is mandatory at or before stage N.

Consider a stationary version of the problem (state and disturbance spaces, disturbance distribution, control constraint set, and cost per stage are the same for all times). At each state x_k and at time k, if we stop, the system moves to a termination state at a cost $C(x_k)$ and subsequently remains there at no cost. If we do not stop, the system moves to state $x_{k+1} = f(x_k, w_k)$ at cost $g(x_k, w_k)$. The terminal cost, assuming stopping has not occurred by the last stage, is $C(x_N)$. An example is a problem of optimal exercise of a financial option where x is the asset's price, $C(x) = x$, and $g(x, w) \equiv 0$.

The DP algorithm (for states other than the termination state) is given by

$$J_N^*(x_N) = C(x_N), \tag{2.70}$$

$$J_k^*(x_k) = \min\Big[C(x_k),\, E\Big\{g(x_k, w_k) + J_{k+1}^*\big(f(x_k, w_k)\big)\Big\}\Big], \tag{2.71}$$

and it is optimal to stop at time k for states x in the set

$$S_k = \Big\{x \;\Big|\; C(x) \le E\Big\{g(x, w) + J_{k+1}^*\big(f(x, w)\big)\Big\}\Big\}.$$

Consider now the rather primitive base policy π, whereby *we stop at every state x*. Thus we have for all x_k and k,

$$J_{k,\pi}(x_k) = C(x_k).$$

The rollout policy is stationary and can be computed on-line relatively easily, since $J_{k,\pi}$ is available in closed form. In particular, the rollout policy is to stop at x_k if

$$C(x_k) \le E\Big\{g(x_k, w_k) + C\big(f(x_k, w_k)\big)\Big\},$$

i.e., if x_k is in the set S_{N-1}, and otherwise to continue.

The rollout policy also has an intuitive interpretation: it stops at the states for which it is better to stop rather than continue for one more stage

and then stop. A policy of this type turns out to be optimal in several types of stopping applications. Let us provide a condition that guarantees its optimality.

We have from the DP Eqs. (2.70)-(2.71),

$$J^*_{N-1}(x) \leq J^*_N(x), \qquad \text{for all } x,$$

and using this fact in the DP equation (2.71), we obtain inductively

$$J^*_k(x) \leq J^*_{k+1}(x), \qquad \text{for all } x \text{ and } k.$$

Using this fact and the definition of S_k we see that

$$S_0 \subset \cdots \subset S_k \subset S_{k+1} \subset \cdots \subset S_{N-1}. \tag{2.72}$$

We will now consider a condition guaranteeing that all the stopping sets S_k are equal. Suppose that the set S_{N-1} is *absorbing* in the sense that if a state belongs to S_{N-1} and we decide to continue, the next state will also be in S_{N-1}:

$$f(x,w) \in S_{N-1}, \qquad \text{for all } x \in S_{N-1},\ w. \tag{2.73}$$

We will show that equality holds in Eq. (2.72) and for all k we have

$$S_k = S_{N-1} = \Big\{ x \in S \;\Big|\; C(x) \leq E\Big\{ g(x,w) + C\big(f(x,w)\big) \Big\} \Big\}.$$

Indeed, by the definition of S_{N-1}, we have

$$J^*_{N-1}(x) = C(x), \qquad \text{for all } x \in S_{N-1},$$

and using Eq. (2.73) we obtain for $x \in S_{N-1}$

$$E\Big\{ g(x,w) + J^*_{N-1}\big(f(x,w)\big) \Big\} = E\Big\{ g(x,w) + C\big(f(x,w)\big) \Big\} \geq C(x).$$

Therefore, stopping is optimal for all $x_{N-2} \in S_{N-1}$ or equivalently $S_{N-1} \subset S_{N-2}$. This together with Eq. (2.72) implies $S_{N-2} = S_{N-1}$. Proceeding similarly, we obtain $S_k = S_{N-1}$ for all k. Thus the optimal policy is to stop if and only if the state is within the set S_{N-1}, which is precisely the set of states where the rollout policy stops.

In conclusion, if condition (2.73) holds (the one-step stopping set S_{N-1} is absorbing), the rollout policy is optimal. Moreover, the preceding analysis [cf. Eq. (2.72)] can be used to show that even if the one-step stopping set S_{N-1} is not absorbing, the rollout policy stops and is optimal within the set of states $x \in \cap_k S_k$, and correctly continues within the set of states $x \notin S_{N-1}$. Contrary to the optimal policy, it also stops within the subset of states $x \in S_{N-1}$ that are not in $\cap_k S_k$. Thus, even in the absence of condition (2.73), the rollout policy is quite sensible even though the base policy is not.

We next discuss a special case of the preceding example. Again the one-step lookahead/rollout policy is optimal, despite the fact that the base policy is poor. Related examples can be found in Chapter 3 of the DP textbook [Ber17a].

Example 2.7.2 (The Rational Burglar)

A burglar may at any night k choose to retire with his accumulated earnings x_k or enter a house and bring home a random amount w_k. However, in the latter case he gets caught with probability p, and then he is forced to terminate his activities and forfeit all of his earnings thus far. The amounts w_k are independent, identically distributed with mean $\overline{w}$. The problem is to find a policy that maximizes the burglar's expected earnings over N nights.

We can formulate this problem as a stopping problem with two actions (retire or continue) and a state space consisting of the real line, the retirement state, and a special state corresponding to the burglar getting caught. The DP algorithm is given by

$$J_N^*(x_N) = x_N,$$

$$J_k^*(x_k) = \max\Big[x_k,\ (1-p)E\big\{J_{k+1}^*(x_k + w_k)\big\}\Big].$$

The one-step stopping set is

$$S_{N-1} = \big\{x \mid x \geq (1-p)(x+\overline{w})\big\} = \left\{x \;\middle|\; x \geq \frac{(1-p)\overline{w}}{p}\right\},$$

(more accurately this set together with the special state corresponding to the burglar's arrest). Since this set is absorbing in the sense of Eq. (2.73), we see that the one-step lookahead/rollout policy by which the burglar retires when his earnings reach or exceed $(1-p)\overline{w}/p$ is optimal. Note that the base policy of the burglar is the "timid" policy of always retiring, regardless of his accumulated earnings, which is far from optimal.

2.7.1 Simplified Rollout and Policy Iteration

The cost improvement property (2.68) also holds for the simplified version of the rollout algorithm (cf. Section 2.3.4) where the rollout policy is defined by

$$\tilde{\mu}_k(x_k) \in \arg\min_{u_k \in \overline{U}_k(x_k)} Q_{k,\pi}(x_k, u_k), \tag{2.74}$$

for a subset $\overline{U}_k(x_k) \subset U_k(x_k)$ that contains the base policy control $\mu_k(x_k)$. The proof is obtained by replacing the last inequality in the argument of Eq. (2.69),

$$\min_{u_k \in U_k(x_k)} Q_{k,\pi}(x_k, u_k) \leq Q_{k,\pi}\big(x_k, \mu_k(x_k)\big),$$

with the inequality

$$\min_{u_k \in \overline{U}_k(x_k)} Q_{k,\pi}(x_k, u_k) \leq Q_{k,\pi}\big(x_k, \mu_k(x_k)\big).$$

The simplified rollout algorithm (2.74) may be implemented in a number of ways, including control constraint discretization/approximation, a random search algorithm, or a one-agent-at-a-time minimization process, as in multiagent rollout.

The simplified rollout idea can also be used within the infinite horizon policy iteration (PI) context. In particular, instead of the minimization

$$\tilde{\mu}(x) \in \arg\min_{u \in U(x)} E_w\Big\{g(x,u,w) + \alpha J_\mu\big(f(x,u,w)\big)\Big\}, \qquad \text{for all } x, \quad (2.75)$$

in the policy improvement operation, it is sufficient for cost improvement to generate a new policy $\tilde{\mu}$ that satisfies for all x,

$$E_w\Big\{g\big(x,\tilde{\mu}(x),w\big)+\alpha J_\mu\big(f(x,\tilde{\mu}(x),w)\big)\Big\} \leq E_w\Big\{g(x,u,w)+\alpha J_\mu\big(f(x,u,w)\big)\Big\}.$$

This cost improvement property is the critical argument for proving convergence of the PI algorithm and its variations to the optimal cost function and policy; see the corresponding proofs in the books [Ber17a] and [Ber19a].

2.7.2 Certainty Equivalence Approximations

As in the case of deterministic DP problems, it is possible to use ℓ-step lookahead, with the aim to improve the performance of the policy obtained through approximation in value space. This, however, can be computationally expensive, because the lookahead graph expands fast as ℓ increases, due to the stochastic character of the problem. Using *certainty equivalence* (CE for short) is an important approximation approach for dealing with this difficulty, as it reduces the size of the ℓ-step lookahead graph. Moreover, CE mitigates the potentially excessive simulation because it reduces the stochastic variance of the Q-factors calculated at each stage.

In the pure but somewhat flawed version of this approach, when solving the ℓ-step lookahead minimization problem, we simply replace *all* of the uncertain quantities $w_k, w_{k+1}, \ldots, w_{k+\ell-1}, \ldots, w_{N-1}$ by some nominal value $\overline{w}$, thus making that problem fully deterministic. Unfortunately, this affects significantly the character of the approximation: when w_k is replaced by a deterministic quantity the Newton step interpretation of the underlying approximation in value space scheme is lost to a great extent.

Still, we may largely correct this difficulty, while retaining substantial simplification, by using CE for *only after the first stage* of the ℓ-step lookahead. We can do this with a CE scheme whereby only the uncertain quantities $w_{k+1}, \ldots, w_{N-1}$ are replaced by a deterministic value $\overline{w}$, while w_k is treated as a stochastic quantity.†

† Variants of the CE approach, based on less drastic simplifications of the probability distributions of the uncertain quantities, are given in the books [Ber17a], Section 6.2.2 and [Ber19a], Section 2.3.2.